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Cadena de Markov

Definición 1 (Cadena de Markov). Sean $\forall n \in \mathbb{N} : X_n \colon \Omega \longrightarrow \mathbb{N}$ variables aleatorias discretas, $(X_n)_{n=0}^{\infty}$ es una cadena de Markov $\iff \forall n \in \mathbb{N} \cup \{0\} : \forall i, j \in \mathbb{N}$:

$$\mathbb{P}\left(X_{n+1} = j \mid X_n = i \wedge \ \ldots \ \wedge X_0 = i_0\right) = \mathbb{P}\left(X_{n+1} = j \mid X_n = i\right).$$

Es decir, la probabilidad de que la variable aleatoria X_{n+1} tome el valor j condicionada a todos los valores anteriores de la cadena, solo depende del valor de la variable aleatoria X_n.

Referenciado en

- `cadena-markov-accesibilidad`
- `cadena-markov-irreducible`
- `cadena-markov-homogenea`
- `cadena-markov-transitoriedad`
- `cadena-markov-comunicacion`
- `cadena-markov-recurrencia`

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